Physicomimetics in 3D with real vehicles

The whole book in Bullet Physics Playground: the plan · 25 October 2026

This file maps every simulation, experiment and checkable result in *Physicomimetics: Physics-Based Swarm Intelligence* (Spears & Spears, Springer 2011) to a Bullet Physics Playground (BPP) port. It covers all 19 chapters and the appendix: 182 entries. Section 5 adds the ideas from the earlier chat, the plan and the thesis digests that go beyond the book. Section 6 extends the capstone mission to ground, surface and underwater vehicles, to show that the method is not tied to one vehicle. Section 7 proposes an escape from cul-de-sacs modelled on the chain fountain. The detailed notes for each chapter are in `book/` (`ch01.md` … `ch19.md`, `ch20-appendix.md`). They give the algorithm, parameter values, the results a port must reproduce (with page, figure and table references), and what the porter must decide where the book is silent or inconsistent.

Page numbers are PDF pages of `apbook1.pdf`. Book page = PDF page − 31.

Status: done = in BPP now; **partly** = in BPP but not yet checked against the book's numbers; **plan** = in `AP-3D-vehicles-plan.pdf`; **new** = not in the plan. A ★ marks an extension that is neither in the book nor in the app.

Order inside each task: port it first as a 2D particle replica and reproduce the book's numbers. Then fly it on vehicles. Then take it to 3D. A 2D replica is what makes a later vehicle or 3D result interpretable.

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1. What reading the whole book changed

These correct or extend the plan. Each is backed by the chapter notes.

1. **Ch. 13 is four mechanisms, not "friction".** In order of cost:
   - **AR damping η** (Eq. 13.9–13.10): scale the force by η = (|D − r|/D)^p_d for r < 2D, then stop the vehicle when |ΣF| falls below a threshold. This is a two-line change and the cheapest jitter fix.
   - **Adaptive friction** (Eq. 13.23), with k_more/k_less > 10.
   - **Static friction** (Eq. 13.25). Its printed condition is dimensionally odd; use μ·m·g < |ΣF·t̂|.
   - **Adaptive speed cap** near obstacles (Eq. 13.24). Its gain ratio is reversed: k_less/k_more > 10.

   The book gives ratios but no absolute gains. Build all four as separate switches (old behaviour at 0) and measure them one at a time. With η and the stop threshold the Pioneers "settle into fixed lattice formations" (PDF 454). Our expectation, not the book's, was that η alone makes the Newtonian drones settle like LJ, because both are then smooth through zero force at D. **Measured (M1, FINDINGS F43–F50): it does,** in 2D and 3D, as fast as LJ. With the book's band (r < 2D) a 2D launch settles at ~0.8 D; damping only within 0.2 D of D (William) keeps D. The stop rule adds nothing without η.
2. **Fixed-wing is not a dumbbell.** §13.5 says a vehicle that can only turn needs a *single* heading particle; that is the torque law of §13.3 (Eq. 13.15–13.18), which the chapter applies to its UAVs. The dumbbell (Eq. 13.19–13.22) is for vehicles that drive forwards and backwards.
3. **"APO from altitude" should build on Ch. 19, not Ch. 18.** Ch. 18's APO is an abstract 50-D optimiser: it collapses onto the best particle and shares fitness all-to-all. Ch. 19's APO is the triangle formation plus a pairwise fitness force (apoF, within 1.5D). It is local, deterministic and the same under rotation, and Ch. 19 calls it "a very simple modification" of the formation code. Sensing from altitude is our own idea (★); neither chapter mentions altitude.
4. **Kinetic theory (Ch. 4.4, Ch. 7) is velocity-based, with no forces.** Nothing in it oscillates about a switch, so it should not jitter on drones. That makes it a good contrast case for Ch. 13. Ch. 7 is entirely 2D, so 3D is our extension.

5. **The evolved chain law (Ch. 12) is nearly always repulsive.** This is our inference, not the book's statement: the evolved minor axis (95.7) exceeds the 60-pixel sensing range, so robots in range almost always repel, and the chain holds together through backflow pulses. Expect it to jitter on drones and to need the Ch. 13 mechanisms.
6. **Ch. 14 is not self-contained.** It does give the offline world (900×700, 40 robots, 90 obstacles, D 50, Vmax 20), the operators (fitness-proportional selection, one-point crossover, Gaussian mutation) and the fitness rule (three equally weighted penalties for collisions, cohesion and reaching within 4D of the goal; Eq. 14.5, PDF 477–480). It does not give the penalty sizes, population size, generations, mutation σ, the Daedalus reward update or a definition of "stuck". The only evolved LJ values in the book are on the Fig. 15.1 screenshot (PDF 519). Their robot–robot force cap (1.08) is about 3.6× below Fig. 14.3's (~3.9), while the obstacle cap (4.32) matches Fig. 14.4. Hettiarachchi's thesis fills some of this (see the thesis digest).
7. **BPP's `math.random` is the generator the appendix warns about; the shared generator replaces it (G2, done).** BPP links Lua 5.1 or 5.2, whose `math.random` calls glibc `rand()`, i.e. `random()`. Run inside Lua 5.1 with Listing A.1's sampling (S extra calls per accepted point, 10^6 points, 36 bins), it fails at S = 25 (χ^2 321), 26 (223) and 27 (409) against a 99.9% cut of 66.6. The Marsaglia multiply-with-carry generator from `Uniform_Coverage.lua` passes at every S tried (χ^2 26–47). It now lives in `lib/rng.lua`, seeded from `SEED` or a `seed` parameter, and Uniform Coverage, the formations, `quadcopter/swarm.lua` and the probes all use it; `checks/rng_appendix.lua` is the regression test.
8. **The Maxelbot sensor is planar.** Its parabolic cones confine sound to a horizontal plane (Ch. 10.5.4.1). A Maxelbot-faithful 3D swarm must either fly in levels or assume a different (4-receiver) sensor. Treat that as a design decision, not an automatic upgrade.
9. **Clustering doesn't explain the 2D Newtonian jitter, but 3D needs a check.** Eq. 3.4 gives the clustering threshold G_t = Fmax·D^p/(2√3) for the triangle, and Eq. 3.5 gives Fmax·D^p/(2+2√2) for the square. The 2D drone defaults are below it (G 1500 against G_t 2368 triangle and 1699 square), so drones don't share nodes. The 3D square uses G 7500, which is above the 2D formula's 5492. A 3D G_t has not been derived, so measure it with a G sweep.
10. **Missing from the plan entirely:**
    - Ch. 2's integrator tests;
    - Ch. 4's Couette flow;
    - Ch. 5's local oriented potential fields (four formations without virtual colours, plus a gripper and a V that changes shape);
    - Ch. 10's sensor and vehicle stack;
    - Ch. 16's heterogeneous swarm design (rings, protection, surveillance, tracking);
    - Ch. 17's two perfect-hexagon methods;
    - Ch. 1's small demos (magnets chain, cooperative pull).
11. **The book's code listings contain bugs; don't copy them.**
    - Ch. 6: the APO sign in Listing 6.3 is reversed relative to Listing 6.4.
    - Ch. 7: speed `max(…, 6)` should be `min`, and Listing 7.3 draws collision angles from a half-circle only.
    - Ch. 3: the Hooke cap applies to the attractive side only.
    - Ch. 13: the signs of Eq. 13.1 contradict the prose.
    - Ch. 18: Theorem 18.1's characteristic equation has a sign typo.
    - Ch. 9: the time units are inconsistent.

    Each chapter note lists its own.

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2. Shared groundwork

Most chapters need the same pieces. Build each once.

ID	Piece	Needed by
G1	**Verification monitors (done, `lib/monitors.lua`):** COM drift, linear and angular momentum, energy. A Lua integrator with controllable Δt for particle tests	2, 3, 4, and every particle replica
G2	**RNG module (done, `lib/rng.lua`):** move the multiply-with-carry generator from `Uniform_Coverage.lua` into a shared module, use it in the formations and the quadcopter harness, add a seed setting (not `os.time()`), and keep the appendix check as a regression test	4, 7, 11, 12, 15, 17, 18, 19, A
G3	**Headless batch runner (done, `lib/batch.py`):** seeds × configs, CSV out, 2 jobs in parallel, no drawing	4, 6, 7, 9, 11, 12, 14, 15, 17, 18, 19
G4	**Velocity commands** for the quad controller (fly this velocity, not this waypoint)	4.4, 7, 11, 13, 17
G5	**Point-mass vehicle with limits:** speed, acceleration and lag as parameters, for large swarms	7, 11, 15, 17
G6	**Scalar fields:** analytic functions, grids and diffusion, drawn as a ground texture or 3D slices	1.5, 6, 8, 9, 10, 19
G7	**Neighbour sensing model:** range and bearing noise, maximum range, token-ring staleness (120 ms × N), 1-bit/packet payload, a Maxelbot preset calibrated to Fig. 10.14	3.4, 6, 10, 11, 12, 15
G8	**Rays:** line of sight between agents, and IR, sonar-ring and laser rangefinders (Bullet ray tests)	5, 7, 9, 10, 11, 12, 13, 14
G9	**Force-law table** keyed by (source type, target type): asymmetric, with wildcards and virtual particles, and one sign convention	3.4, 3.5, 5, 16, 17
G10	**Ground vehicles (Pioneer 3-AT done, `ground/`; Maxelbot to come):** skid-steer Maxelbot (MMP-5) and differential-drive Pioneer 3-AT	9, 10, 11, 12, 13
G11	**Water vehicles (the ASV done, `boat/`):** a holonomic round ASV (Voith–Schneider); an ROV-style multi-thruster vehicle that can hover and move sideways; a torpedo-shaped AUV that cannot	6, ★9, ★capstone
G12	**Fixed-wing aircraft (kinematic model done, `fixedwing/`; Ch. 13 experiments in `ch13/`):** constant speed, bank limit, with the heading-particle torque law	3.1 ★, 13
G13	**Worlds:** wall polygons from vertex lists, corridors, mazes, and an obstacle-density generator	4.3, 7, 11, 12, 14, 15
G14	**Plume model:** Farrell-style puffs with wind, white-noise mixing and obstacles	8, 9
G15	**Neighbour messages:** sensed values as well as positions; flooding broadcast	4.1, 6, 9, 12, 19
G16	**Metrics (lattice quality started, `lib/lattice.lua`; drone settling in `probes/settle_drones.lua`):** lattice quality (angles, Delaunay CV), coverage grids and KL divergence, chain metrics (MST), suggested-versus-actual position gap, energy and path length, Q versus P	all
G17	**Heading-dependent force terms:** elliptical zones, LOPF sectors	4.2, 5, 12
G18	**One interface for every vehicle (done, `lib/swarm.lua`: the quadcopter, the ideal vehicle `lib/ideal.lua`, the ground robot and the boat; the laws shared in `lib/laws.lua`):** physicomimetics outputs a desired velocity; an adapter per vehicle turns it into commands. Vehicles that cannot move sideways use Ch. 13's heading particle or dumbbell	13, ★capstone, every vehicle port
G19	**Water:** the surface as a boundary, buoyancy, drag and currents (forces in Lua; BPP has no fluid simulation), and a plume carried by the current	6, ★9, ★capstone
G20	**Underwater sensing and comms:** sound only, ~1500 m/s, delayed and low-bandwidth; no GPS or radio below the surface; surface boats as gateways between sound and radio	★6, ★capstone

ID	Piece	Needed by
G21	**World generator (core and book presets done, `lib/world.lua`; realistic layers done: `lib/terrain.lua`, `lib/town.lua`, `lib/buildings.lua`, `lib/features.lua`, `lib/vegetation.lua`, with the five realistic presets in `lib/places.lua`; wind and currents later):** our own seeded, parameterised 3D generator, modelled on real places: terrain, water with depth, zones from rural to city centre, streets, lots, buildings of varied footprint and floor count, vegetation, and the capstone's harbour features. Presets: rural, suburban, city centre, industrial port, harbour town, and the book's worlds. Nothing from the CPT map makers; it writes Zarzhitsky's map format only so his plume tools can check our port (§6)	3, 4, 7, 9, 11, 12, 14, 15, ★§7, ★capstone

3. Overview by chapter

Ch.	Title (author)	Entries	First BPP target	Vehicles	Groundwork	Status
1	Nature is lazy (Spears)	8	Potential-well height map; Q-vs-P metric	particles, ★ magnets	G6, G3	new
2	NetLogo and physics (Spears)	3	Springs and solar system as integrator tests	particles	G1	partly
3	NetLogo and physicomimetics (Spears)	7	Check the existing formations against the book; add Hooke, honeycomb, events	particles, quads	G1, G9, G16	partly
4	Pushing the envelope (Spears)	5	Couette flow (2 closed forms); perfect square	particles, quads	G1, G3, G4, G15, G17	partly / plan / new
5	Local oriented potential fields (Bravi et al.)	9	Square, line, star, tree by sector forces	holonomic particles, quads	G9, G17	new
6	Bioluminescence (Karlsson et al.)	7	Slime-mould field + APO swarm, min and max tasks	particles, ASV, ★ AUV	G3, G6, G11, G15	plan
7	Gas-mimetic swarms (Kerr)	6	Three theory checks, then KT vs Default vs Ant	particles, quads, ground robots	G3, G4, G8, G13	plan
8	Fluxotaxis theory (Spears, Thayer, Zarzhitsky)	6	3D closed-form unit tests	—	G6	plan
9	Fluxotaxis experiments (Zarzhitsky)	14	Puff plume + 7-robot and swarm fluxotaxis	particles, Maxelbots, quads	G3, G8, G14, G15	plan
10	What is a Maxelbot? (Maxim)	13	Trilateration noise model + token ring	Maxelbot	G7, G8, G10	new
11	Uniform coverage (Maxim)	13	KL metric, wall-only control, f = 0.6	particles, Maxelbot, quads	G3, G13, G16	partly
12	Chain formations (Maxim)	18	Listing 12.4 maze with the evolved law	particles, Maxelbots, quads	G3, G7, G8, G13, G17	plan
13	Motion control of constrained agents (Apker, Potter)	13	η, adaptive and static friction on the drones	quads, Pioneer, fixed-wing	G4, G10, G12	partly (η on the drones; the Pioneer measured)
14	Adaptive learning, Daedalus (Hettiarachchi)	7	Offline three-law LJ obstacle course	particles, quads	G3, G8, G13	plan
15	Statistical scouts (Spears et al.)	10	Bernoulli vs Bayes on the Ch. 14 world	particles (drones for show)	G2, G3, G13	plan

Ch.	Title (author)	Entries	First BPP target	Vehicles	Groundwork	Status
16	Swarm design considerations (Wiegand, Ellis)	9	Type table; rings; protection	particles, quads	G9, G16	new
17	Swarm engineering (Kazadi)	9	Hexagon Methods 1–2; quark model	particles, quads	G2, G3, G4	plan / new
18	APO for global optimisation (Xie, Tan, Zeng)	8	Table 18.2 as a headless benchmark	—	G2, G3	new
19	AP for noisy nonstationary environments (Spears)	13	apo.nlogo demo, N × D table, tracking	particles, quads	G3, G6	plan
A	Anomalous random() (Spears, Green)	4	RNG regression test and demo	—	G2	partly

4. Every entry, chapter by chapter

Each row lists what the entry becomes in BPP. The chapter note has the full detail.

Ch. 1 — Nature is lazy (PDF 34–57) · `book/ch01.md`

#	Entry	Pages	BPP	Status
1.1	Swarm baseline P = 1 − Π(1 − p_i) versus measured Q	37–38	Standard summary line in the batch runner (G3)	new
1.2	Cooperative pull (video)	38	Ground robots or ★ quads with tethers moving a payload	new
1.3	Magnets self-organise into a chain (video; 39 + 1 dropped)	42	Rigid discs with dipole force and torque	new ★
1.4	Compass construction of a triangular lattice	44–45	Ground-truth placement; ★ 3D sphere intersections	new
1.5	Rubber-sheet potential wells, incremental and batch	45–48	Live potential height map under 7 particles (G6)	new
1.6	MAV hexagonal sensing grid	40	Done as the drone triangle; add a grid-quality score	done
1.7	Least action	41–44	Energy and path length as standard metrics (G16)	new

Ch. 2 — NetLogo and physics (PDF 58–85) · `book/ch02.md`

#	Entry	Pages	BPP	Status
2.1	`spring1D` : Hooke oscillation, momentum 0, energy flat, friction drains it	58–72	`ch02/` : scene and checks, all results reproduced	done
2.2	`spring2D` : angular momentum conserved, centrifugal stretch > 2D	73–77	`ch02/` : scene and checks, reproduced; ★ 3D off-plane spin; ★ spinning tetrahedron (tennis-racket flips); ★ 3D spring lattice	done
2.3	`solar_system` : Δt failures, velocity cap fails, force cap works (precession)	78–85	`ch02/` : scene and checks, all five experiments reproduced (peak speeds 5.5, 5.0, 4.0, 9.0 as printed); ★ several planets on tilted elliptical orbits, stable with a gap of 1.8 between orbits and ~50 steps per nearest-point pass (F58)	done

Ch. 3 — NetLogo and physicomimetics (PDF 86–123) · `book/ch03.md`

#	Entry	Pages	BPP	Status
3.1	Boids (`flock`): turn caps 5/3/1.5°; snapshots at 546, 1,309, 78, 42 steps	86–95	Constant-speed particles; ★ 3D boids on quads or fixed-wing	plan
3.2	Hooke lattice (`formation_hooke`): k 0.5, Fmax 1; fails at Fmax 3 and 5; kill and birth	96–103	`ch03/models.lua` : the lattice by 200 steps and every failure reproduced (`checks/ch03.lua` , F75); drones to come	partly
3.3	Split-Newtonian triangle, G_t = Fmax·D^p/(2√3), hysteresis, 50 → 21 nodes	103–108	Done; G_t, hysteresis and robustness confirmed on the model and the BPP scene; 3D G_t = G_t/√2 derived and measured (`ch03/` , F73)	done
3.4	Square with two types, defect removal by noise, 500 particles	108–113	Done; Eq. 3.5 fits the Java/BPP force direction, not the .nlogo's (F74); add annealing, type flips, toggling, a quality metric; ★ NaCl in 3D	partly
3.5	Honeycomb: √3 renormalisation, 120°/60° repair, square↔honeycomb toggle	113–115	`ch03/models.lua` (`honeycomb.nlogo` line for line): centres trap particles, the model's switching (a random mix of the two rules per pair, F77) frees them; McLurkin's repair not given in the text; drones at one altitude, ★ diamond lattice to come	partly
3.6	Goal and obstacles; Disable One; abandon threshold goalF > Fmax	115–121	Done; Figs. 3.24–3.26 reproduced on the model; the threshold is exact for a lone neighbour, a bound in a formation (F75)	done
3.7	Hardware requirements (GPS, IR, trilateration)	121–123	Sensing model (G7, G8)	new

Ch. 4 — Pushing the envelope (PDF 124–157) · `book/ch04.md`

#	Entry	Pages	BPP	Status
4.1	Perfect formations (`formation_perfect`): (m, n) slots; the triangle breaks Newton's third law	124–130	Square first (momentum-clean), then triangle; ★ (m, n, k) cube; the author's 3 open questions	plan
4.2	Chains (`formation_chain`): elliptical law; Table 4.1 (561 → 350)	131–138	Particles with heading (G17), 50-run batches; ★ prolate spheroid in 3D	plan
4.3a	Coverage baseline (`uniform`): 84×84, 9 cells; Table 4.2	139–144	Existing particle port; check against Table 4.2	partly
4.3b	Mean free path (`uniform_mfpl`): bound 17⅓; Fig. 4.19 dip at 53–65	144–147	Check against Fig. 4.19; ★ 27 cells in 3D	partly
4.4	Couette flow (`kt`): slope V/6 with r > 0.999; ν̄ = 1.2617√T (theory 1.2533)	147–156	Particles (two closed-form tests) → point mass → quads; ★ 3D slab	new

Ch. 5 — Local oriented potential fields (PDF 160–175) · `book/ch05.md`

#	Entry	Pages	BPP	Status
5.1	Square deployment by 8 sector forces (no virtual colours)	164–165	Particles → drones; compare defects with Ch. 3's square	new
5.2	Linear deployment	165	Same machinery	new
5.3	Star with leader, arms and levels	166	Same; ★ 3D star with 4 or 6 arms	new
5.4	Star "gripper" (arms closing by ±γ)	171	★ Drones grasping a rigid payload	new
5.5	Tree / V with levels	167	★ Flying V of drones	new
5.6	V that changes shape (α changed at run time)	171	Same	new
5.7	Matarić behaviours in AP (wander, follow, disperse, aggregate, home, angularity)	162–168	A behaviour library on the formation code	new
5.8	Holes and choice of starting formation	171–173	Measure holes against the starting lattice	new
5.9	"Angle of attention" (occluded view)	174	Ray-cast occlusion share per formation (G8)	new

The chapter gives no numbers, so the port defines its own parameters and metrics.

Ch. 6 — Bioluminescence (PDF 176–223) · `book/ch06.md`

#	Entry	Pages	BPP	Status
6.1	Dinoflagellate field generator (slime mould, 290/400/510 dinos)	183–188, 200	Grid field + drawing (G6); ★ 3D volume, currents	plan
6.2	ASV swarm: split-Newtonian + APO + goal (only drones with neighbours feel the goal)	188–192	Particles → ASVs	plan
6.3	Force balance: goalF < 0.2598, apoF < 0.13	196–198	Batch sweeps	new
6.4	Max/min tasks: rising H1, U-shaped H2 with N = 6 best; time curves; 1295/1313/1292 ticks with no chemical	198–209	~12,000 particle runs (G3); then ASVs	plan
6.5	Multi-drone simulator (asynchronous clocks, comms, fuel)	210–218	Our harness plus G7, G15 and fuel accounting	plan
6.6	20-drone plume demo	216–217	Demo	plan
6.7	Holonomic ASV hull (18″, two Voith–Schneider propellers)	218–222	`boat/vehicle.lua` (G11): the book's hull, mass and thrust chosen; follows AP like the drones (F67); ★ AUV next	done

Ch. 7 — Gas-mimetic swarms (PDF 224–253) · `book/ch07.md`

#	Entry	Pages	BPP	Status
7.1	KT controller: Rayleigh/Gaussian wall rebound, 15″ collision zone, 24-sonar ring	228–236	Particles → velocity-commanded quads (G4, G8)	plan
7.2	Uniform density at rest, error < 10%	239–240	Batch	plan
7.3	Mean speed versus T, error < 2%	240–241	Batch; ★ 3D Maxwell √(8kT/πm)	plan
7.4	Profiles with one moving wall (linear) and two (flat), against obstacle density	241–243	Batch; decide one- vs two-sided explicitly	plan
7.5	Default and Ant baselines	244–246	Particles; pheromone grid	new
7.6	KT vs Default vs Ant: KT covers ~90% in about a third of Ant's time	246–251	1000″×5000″ corridor, 50×250 grid; ★ 3D canyon or tunnel	plan

Ch. 8 — Fluxotaxis theory (PDF 254–281) · `book/ch08.md`

#	Entry	Pages	BPP	Status
8.1	Task split; the swarm as a finite-difference grid	256–264	Design note	plan
8.2	Emitter test: positive outward flux for time τ	267–268	Termination rule for Ch. 9	new
8.3	GDMF navigation law	269–271	Core law	plan
8.4	Anemotaxis overshoots; fluxotaxis stops at the emitter	272–273	Analytic test	new
8.5	Continuous emitter: GDMF $\propto r^{-3}$ against gradient $\propto r^{-2}$	275–277	3D closed-form unit test	new
8.6	Single puff: gradient → 0 at the emitter, GDMF → ∞	277–280	3D unit test (the note re-derives the garbled formula)	new

Ch. 9 — Fluxotaxis experiments (PDF 282–331) · `book/ch09.md`

#	Entry	Pages	BPP	Status
9.1	1 versus 3 Maxelbots on ethanol: 22% vs 60% success	285–286	Small hardware replica (G10, G14)	new
9.2–9.3	Smoke around a robot; 32-sensor flume	287–288	★ downwash stirring the plume	new
9.4	Finite-difference stencils	291	Library	plan
9.5	Puff plume model	291–295	G14, the largest single build. Zarzhitsky's CPT code (PlumeGen, TraceGen; kept outside the repo) gives the constants the book omits; the Android app's Chemical Plume Tracing is ported from it with a changed generator. Run the original on our generator's maps as an independent check	plan
9.6	LJ variant F = D/r² − D^1.7/r^2.7	296–299	Add to the force-law table	new
9.7	Simulator loop, collision avoidance overrides tracing	297–300	Harness	plan
9.8	7-robot hexagon: casting, chemotaxis, anemotaxis, fluxotaxis	300–306	Particles → quads	plan
9.9	Swarm versions (45° casting, flux toward each neighbour)	306–312	Unchanged in 3D: the cheapest route to 3D	plan
9.10	Exp. 1: success 74.6% fluxotaxis (Table 9.3)	312–315	Batch	plan
9.11	Exp. 2: obstacles, emitter enclosed 49.1% of the time	315–318	Batch	plan
9.12	Exp. 3: 7–70 robots, 40,500 runs (Tables 9.4–9.5)	318–323	Batch (subset)	plan
9.13	Exp. 4: 10–150 robots (Table 9.6)	323–326	Batch; ranking reverses against 9.12 (book inconsistency)	plan
9.14	Urban demo	328–329	★ 3D city with source height	plan

Ch. 10 — What is a Maxelbot? (PDF 332–371) · `book/ch10.md`

#	Entry	Pages	BPP	Status
10.1	Closed-form trilateration (6″ L)	333–335	Sensor model core; ★ 4-receiver 3D version	new
10.2	Multi-hop coordinate sharing	335	G15	new
10.3–10.5	SRF04 accuracy; wavefront jitter fixes; range 2 → 4 → 9–11 ft	336–344	Noise and range parameters	new
10.6	Parabolic cones: sensing is planar	345–346	Elevation limit in 3D (decision)	new
10.7–10.8	RF Manchester link; module timing	345–351	Latency and drop model	new
10.9	Grid test: mean error 0.6″, max 1.2″; ≤ 11% at 9–11 ft	351–352	Calibration target for G7	new
10.10	Token ring, 120 ms per slot, recovery	352–355	Staleness scheduler; ★ how large N can grow at drone speeds	new

#	Entry	Pages	BPP	Status
10.11	One robot moves at a time (centralised design)	356–357	★ Sequential update mode as a jitter test	new
10.12	Localisation I/O (x, y, heading byte)	357–361	Neighbour-table format	new
10.13	MMP-5 skid-steer base, IR 10–80 cm, compass, temperature; 26″ follower	361–370	Ground vehicle (G10); ★ heat-source follower	new

Ch. 11 — Uniform coverage (PDF 372–397) · `book/ch11.md`

#	Entry	Pages	BPP	Status
11.1–11.3	Maxim's simulator, cell tiling, KL metric (natural log)	373–376	Metric (G16)	partly
11.4	Markov-chain existence proof	376–377	Analytic test	new
11.5	Turns at walls only: centre least visited (negative control)	378–379	Particles	new
11.6	Mean-free-path turns at f = 0.6 × cell: KL 0.000060	379–381	Particles (ported, check KL)	partly
11.7	Mean free path theory 0.6518 (0.6079 with buffer)	380–383	`ch11/`: 2D reproduced (exact 0.651757, buffer 0.6079); ★ 3D 0.7720 × edge (buffer 0.7138), with a closed form	done
11.8	12 environments × 5 cell sizes (Table 11.1)	383–386	G13 (most layouts must be rebuilt)	new
11.9	Multi-robot formula 1 − (1 − 1/C)^N	384	Check with N drones ★	new
11.10a–c	Hardware runs: 5×5 ft, 5×15 ft, L-shape	384–396	Maxelbot replica (G7, G10)	new
—	3D coverage, altitude bands, per-minute scoring	—	★ from the plan	plan

Ch. 12 — Chain formations (PDF 398–443) · `book/ch12.md`

#	Entry	Pages	BPP	Status
12.1–12.3	Elliptical law, backflow force, front tangential force	399–402	Particles (G17)	plan
12.4	EA with 11 evolved parameters (Listing 12.2)	402–404	Use the published values	plan
12.5–12.7	Maxelbot simulator, 35-vertex maze (Listing 12.4), metrics, 9 environments	404–413	G13, G16	plan
12.8–12.10	Exp. I scalability, II noise, III add then kill	414–418	Batch, 10 seeds	plan
12.11	35-step backflow pulse test	419	Check	new
12.12	Ideal chain length (closed form)	419–421	Check (the book's example gives 37; the formula gives 38.5)	new
12.13–12.16	Chain-list protocol, display, laptop link, token loss	422–428	G15; ★ chain as a comms relay	new
12.17	Static Maxelbots: line, L, U at 2, 4, 6 ft	429–436	Replica (G10)	new
12.18	Dynamic: open floor, 5×14 ft corridor, L corridor	436–442	Replica; then ★ 3D shafts on quads	plan
★	Chain-fountain escape (§7): the same idea of a chain, used to pull agents out of a trap rather than to deploy a relay	—	Compare with the Ch. 12 law	new

#	Entry	Pages	BPP	Status
★	Chain applications: relays in tunnels, mines, caves, garages, urban canyons, stairwells, over buildings, into pits; maze exploration; map carried back to base; branching at junctions; loops that reconnect; wall maps from range hits. Score: does the line-of-sight link hold?	—	Quads after the 2D replica (see §5)	new

Ch. 13 — Motion control of constrained agents (PDF 444–471) · `book/ch13.md`

#	Entry	Pages	BPP	Status
13.1–13.2	Point-mass reachable space; the discontinuity at D	447–448	Explains the Newtonian jitter	done (observed)
13.3–13.4	Differential-drive and Pioneer reachable space	449–453	`ground/vehicle.lua` (G10): the Pioneer with the book's r_diff, wheel range and caps; read as stopping the wheel, the minimum speed freezes the lattice as Fig. 13.7 predicts (F66); read as turning it at 0.1 m/s, the robots reach D and get through the obstacle field (F72); plots to come	partly
13.5	**AR damping η + stop threshold**	453–454	`lib/ch13.lua`; on the drones `etaPower` 1, `etaWidth` 0.2 (2D) / 0.5 (3D), the Newtonian default; measured 2D/3D against LJ (`quadcopter/README.md`, F43–F55)	done
13.6	"Triangle of death"	454–456	Ground robots	new
13.7–13.8	Fixed-wing reachable space (13 m/s, 20° bank); single-attractor behaviours. Eq. 13.14 gives a ~101 m turn radius, which matches Fig. 13.16's loops; a coordinated turn would give ~47 m	457–459	`fixedwing/model.lua`, `ch13/UAV-Attractor.lua`: the three outcomes of Fig. 13.13 reproduced (F76)	done
13.9	Heading-particle torque law	460–462	`ch13/uav.lua` on the fixed-wing (Eqs. 13.15–13.18; repulsion turns away from the neighbour, F76)	done
13.10	Dumbbell agent	462–463	Pioneer	plan (corrected)
13.11	**Pioneer recipe: adaptive friction (13.23), caps, adaptive cap (13.24), static friction (13.25); 3–5-robot camera ring with three-point turns**	463–465	On the drones all measured: only η is needed; adaptive and static friction and the adaptive cap are not (F51, F54, F55). On the Pioneers (M2): adaptive and static friction do not settle the split law at D, and the cap has no collisions to prevent (F66, F68); ring on Pioneers to come	partly
13.12	5 UAVs and 10 targets, gravity vs torque law	465–467	`ch13/UAV-Targets.lua`, 20 seeds: the torque law 1.6× the time over the targets and a tighter spread; not the lower turning (F76)	done
13.13	Four-step design procedure	466–468	Checklist	new

Ch. 14 — Adaptive learning, Daedalus (PDF 472–505) · `book/ch14.md`

#	Entry	Pages	BPP	Status
14.1	Offline EA for three LJ laws (Table 14.2 target)	475–484	Particles; values from Fig. 15.1	plan
14.2	50 robots flow round a large obstacle	483	Particles → drones	partly
14.3	Offline law in the 5-goal corridor: 51.66% survive	485–488	Batch	plan
14.4	Line of sight between robots	489–490	G8	plan
14.5	Mutation-rate groups, Approaches 1 and 2 (48.3 → 68.3%)	491–497	Daedalus loop	plan

#	Entry	Pages	BPP	Status
14.6	Survival against obstacle density (Tables 14.7–14.10)	497–500	Batch, the cleanest numeric target	plan
14.7	Future: stuck robots switch to the gas model	503	★ LJ → KT hybrid escape; compare with the chain-fountain escape (§7)	new
★	Chain-fountain escape: a free agent leads stuck agents out of a cul-de-sac as a chain	—	§7	new

Ch. 15 — Statistical scouts (PDF 506–535) · `book/ch15.md`

#	Entry	Pages	BPP	Status
15.1–15.2	Bernoulli and Bayes formulas (7 priors; closed form)	511–517	Library	plan
15.3	Navigation simulator: 800×600, 1,500 steps	517–519	The Ch. 14 world	plan
15.4	Ground truth from 1,000 runs, MSE	519–524	Batch (G3); environments with p = 0.1/0.5/0.9 to design	plan
15.5–15.8	Hypotheses 1–4 (prior, more scouts, scalability, difficulty)	524–531	Batch	plan
15.9	LJ scouts predict better than Newtonian scouts	532	Batch, both laws	new
15.10	Porting to real robots	533	★ cascade crashes on drones	plan

Ch. 16 — Swarm design considerations (PDF 536–559) · `book/ch16.md`

#	Entry	Pages	BPP	Status
16.1	Split-Newtonian law with a mass exponent	540–541	Force-law table (G9)	new
16.2	Two types make rings of any radius	541–542	Particles → drones; ★ spherical shells	new
16.3	Virtual particles	542	G9	new
16.4–16.5	Asymmetric type-to-type laws; graph-based design	542–550	G9	new
16.6	Resource protection (3 inner + 6 outer)	543–552	Demo; ★ drones guarding a VIP	new
16.7	Multi-field surveillance	545–552	Demo	new
16.8	Covert blind-spot tracking	546–553	Demo	new
16.9	Model-vs-vehicle metrics: suggested-vs-actual gap, settling time, lattice CV	554–557	**Acceptance metrics for the Ch. 13 work**	new

Ch. 17 — Swarm engineering (PDF 560–592) · `book/ch17.md`

#	Entry	Pages	BPP	Status
17.1–17.3	Method; hexagon energy; single hexagon is the minimum	560–572	Theory and checks	new
17.4	Basic run: energy plateaus with defects	572–574	Particles	new
17.5	Method 1: synchronised drift	575–577	Particles	new
17.6	Method 2: edge-agent kicks, hexagon in every one of > 10,000 runs	578–580	Particles; ★ 3D; compare with Ch. 4.1	new
17.7–17.8	Lossless flocking; the informed-agent bias limit (Eq. 17.24)	580–585	Sweep the goal-force share on existing formations	new

#	Entry	Pages	BPP	Status
17.9	Quark model: coalescence 81%, informed chase, squeeze through a hole	586–590	Particles → velocity-commanded quads (G4)	plan

Ch. 18 — APO for global optimisation (PDF 596–620) · `book/ch18.md`

#	Entry	Pages	BPP	Status
18.1–18.2	APO framework; APO1/2/3 laws	600–606	Headless Lua	new
18.3	50-D benchmark, six functions (Table 18.2)	606–609	Batch, plus a control with the optimum shifted from the origin	new
18.4	Seven mass functions	609–612	Batch	new
18.5–18.6	Convergence theorem; choosing G	613–616	Closed-form checks	new
18.7–18.8	EAPO, VM-APO, LAPO; applications (no data)	598, 616–618	Notes	new

Ch. 19 — AP for noisy nonstationary environments (PDF 622–644) · `book/ch19.md`

#	Entry	Pages	BPP	Status
19.1	`apo.nlogo`: 7 particles find the Rastrigin minimum	624–626	Particles + G6	plan
19.2	Noise 600,000 and 5,000,000	626–627	Same	plan
19.3	N × D table (Tables 19.1–19.2)	627–628	**Main numeric target**	plan
19.4	Zoom: feature size against D	627–629	Same	new
19.5	Click to move the optimum (tracking)	629–630	★ drones tracking a moving hotspot	plan
19.6	d + 1 minimum particles	629–630	4 drones in 3D	new
19.7–19.9	APO vs PSO 2D: stationary, moving, rotated ellipse	631–637	Batch; PSO in Lua	new
19.10–19.13	30-D trajectories; PSO → APO hybrid, hand-set and cyclic D	637–643	Headless	new
—	★ Sensing from altitude (blur with height), 3D field	—	Our extension of 19.1–19.5	plan

Appendix — Anomalous random() (PDF 646–652) · `book/ch20-appendix.md`

#	Entry	Pages	BPP	Status
A.1	Angle histogram with S extra calls (S = 25, 27 biased; ≥ 28 clean)	646–648	`checks/rng_appendix.lua` (G2): `math.random` biased at S = 3, 25, 26, 27; `lib/rng.lua` passes at every S	done
A.2	Spatial histogram in the disk	649–650	Visual demo: math.random vs the new RNG	new
A.3	Quarter-circle area (unaffected)	651–652	Control	new
A.4	drand48 and others are fine	651–652	Choice of generator	new

5. Ideas from the earlier chat, the plan and the thesis digests

These come from the conversation that produced `AP-3D-vehicles-plan.pdf`, from that plan, and from the thesis digests' sections on what they mean for the plan. Each one is missing from, or only touched on, in the tables above. They are grouped by the chapter they attach to. Source: **chat** = suggested in the earlier conversation; **plan** = a ★ item in the plan PDF; **digest** = a thesis digest (author named).

Formations (Ch. 3, 4, 17)

Idea	Source
★ **Lag acts like heat:** measure how far drone lag shifts the G phase transition (Eq. 3.4), so the drone port gives a result as well as a demo	plan
★ **Gentle annealing on drones:** random kicks that start large and cool to zero, to shake 3D lattices out of glassy arrangements. On drones the kicks are real accelerations, so keep them small	chat
★ **Choose N to suit the shape:** 13, 55 or 147 for icosahedra, or a count that completes whole hexagonal layers	chat
★ **Three-label spin for FCC:** flight levels give ABAB (HCP) stacking; a three-label version of the spin could force ABC (FCC)	chat
★ **Honeycomb's switching costs energy:** the square↔honeycomb switching that escapes the centre trap keeps drones moving. Measure the trade-off between avoiding the trap, energy and wobble	plan
★ **Diamond caveats:** diamond is floppy under distance-only forces, so it is an open question whether the like-class links hold it rigid; the square-escaping switch needs rethinking, because the competing 3D arrangements differ	chat
★ **Honeycomb at one altitude as ground coverage** for cameras or radio: the most area per drone for a given link distance	chat
★ **Perfect Formation on drones:** the momentum checks show how much tracking lag disturbs the formation; test self-repair when switching formations; "One is Born" mid-flight becomes a launch problem; the shared compass fixes the cube's random rotation about the vertical	chat
★ **Ch. 13 in 3D:** the chapter covers only wheeled robots and fixed-wing aircraft; quadcopters in 3D extend it, and its reachable-space analysis needs a second angle (climb rate)	chat
Launch new drones from the side, so a one-at-a-time launch doesn't climb up through the levels already formed	chat (open item)

Optimization and fluxotaxis (Ch. 8, 9, 19)

Idea	Source
★ **Missions for APO on drones:** finding a radiation hotspot or a gas source	chat
★ **Readings from where the drone actually is,** not from its waypoint; downwash stirring a real plume as a noise source	chat
★ **3D plumes:** buoyant gases rise and heavy ones hug the ground; puffs have height and reflect off the ground	chat
★ **Wind changes with height** (slower near the ground); buildings have heights that make wakes and street canyons. Add a height-dependent mean wind as a 3D boundary condition with spin-up, and keep the obstacle mass-relocation rule against building walls	chat; digest (Zarzhitsky)
★ **Source height:** a swarm spread over several altitudes finds the plume's vertical structure; a rooftop vent and a ground spill look the same in 2D. Report source height as a metric by repeating the main tables with the emitter raised	plan; digest (Zarzhitsky)
★ **Wind from tilt and thrust** as a non-intrusive anemometer (fluxotaxis never ran on hardware for lack of one). Model its lag and noise; it may favour fluxotaxis or chemotaxis over anemotaxis	plan; digest (Zarzhitsky)
★ **Downwash on the sensors:** bias each drone's sensed wind and concentration by its own rotor wash and compare with ideal sensors; test sensors mounted above and below the body (placement mattered on a single robot)	plan; digest (Zarzhitsky)
Pseudo-source test: buildings that create corner sinks or concentration pockets, where chemotaxis is fooled and fluxotaxis should not be	digest (Zarzhitsky)
Flux stopping rule over a 3D shell (sustained outflux for τ), which the thesis never enabled; the natural way to estimate source height	digest (Zarzhitsky)
Re-derive G_t for 3D packing (the thesis notes a 308 vs 231 discrepancy in its Eq. 4.7)	digest (Zarzhitsky)
Compare directly with the app's 2D results: success rate, time to find the source, distance flown; draw the plume as translucent puffs	chat
Share the field code: optimization and fluxotaxis both find a source in a field with a sensor term on top of the lattice force, so the 3D field sensing and drawing carry from one to the other	chat

Bioluminescence (Ch. 6)

Idea	Source
★ **Wake-triggered glow:** real bioluminescence is triggered mechanically, so the vehicles' own wakes light the water. Speed against stealth becomes a control problem	plan
★ **Dinoflagellates migrate vertically** by day and night, and currents add drift to everything. A 3D field of roughly 60 × 60 × 20 cells	plan; chat
★ **Two underwater vehicles:** a multi-thruster ROV-style vehicle (like a BlueROV2) that behaves like an underwater quad, and a torpedo-shaped AUV that cannot hover or move sideways, the harder and more realistic case	chat
Draw the glow with translucent slices or point sprites	chat
Underwater sensing: sound at ~1500 m/s; the RF sync pulse won't propagate, so use an optical or clock-synchronised trigger	digest (Kunkel, Heil)

Coverage and kinetic theory (Ch. 4, 7, 11)

Idea	Source
★ **Volume coverage** (air sampling, searching a warehouse): in 3D the collision cross-section becomes an area, not a width	chat
★ **Does the linear profile survive collisions that take time?** A quad needs half a second or more to change velocity, which should act like extra viscosity	plan
Lag against collision time: log commanded-vs-actual velocity lag (time to ~90%) relative to the mean time between collisions; the profile should hold only while the lag is much shorter. Sweep a controller gain to find the failure point	digest (Kerr)
Track the backward-moving fraction ($v_y < 0$), which Kerr blames for profile error, and the collision rate per bin	digest (Kerr)
More 3D checks: profile uniform in z; $\langle \frac{1}{2}mv^2 \rangle = (3/2)kT$; the full Maxwell speed histogram with a KS test (never tested); altitude bias from floor and ceiling	digest (Kerr)
★ **Fixed-wing drones as gas molecules** (constant speed). Kerr's digest warns that equal speeds break the Maxwell speed distribution, so test the heading distribution instead	plan; digest (Kerr)
Scale: full quads manage ~100–200 in BPP against the app's 1,000 particles, so expect noisier histograms or use the point-mass vehicle (G5)	chat

Chains (Ch. 4.2, 12)

Idea	Source
★ **Where relays matter:** tunnels, mines, caves, parking garages, urban canyons; up a shaft or stairwell, over a building, down into a pit	chat
★ **Chains for exploring maze-like spaces,** not only for holding a link	chat
★ **"Does the relay hold?" as the score:** a link breaks when walls block the line of sight, not only on distance; the app's Max Distance gauge becomes a radio-range check	chat
★ **Heading for a hovering drone:** the ellipse's axis comes from yaw or from a smoothed travel direction. It may decide whether chains stay straight or fold	plan
3D chain details: front tangential force in a random direction perpendicular to the heading; up/down IR sensors for shafts; acoustic line of sight blocked by walls while RF is not	digest (Maxim)
Reproduce the 2D Maxelbot runs first: 12 in/s, 3 IR sensors at 9″ range, 5 ft maximum spacing, the 14 ft and 8 + 12 ft corridors	digest (Maxim)

Learning and scouts (Ch. 14, 15)

Idea	Source
★ **Drones tune their own settings mid-mission** to cut losses on the obstacle field	chat
Report "lost" (crashed) separately from "hit": a crashed quad cannot be repaired at the goal. Repeat the safety-zone result (zero collisions, lower reachability)	digest (Hettiarachchi)

Idea	Source
Train easy, test hard: the online world (obstacles ×3, > 16% coverage, Vmax × 1.5, 5 chained goals) is a ready difficulty knob for our field	digest (Hettiarachchi)
Minimal Daedalus: 13-float genome, worthiness 1000, rule copying on contact within 1.5R and in line of sight, 5% "panic" mutation when stuck, seed rates {1, 3, 5, 7, 9}%; feed the LJ output to the controller as a velocity or acceleration setpoint	digest (Hettiarachchi)
Hardware Hooke repulsion (IR, range 30, front-arc sensors) as a template for realistic obstacle sensing	digest (Hettiarachchi)
Scouts in two modes: independent and in formation (same LJ as the swarm); test whether crash cascades break the thesis's "dependence helps" result	digest (Rebguns)
Cost: ground truth at ~1,000 runs per field is ~17 h with full drones; use G5 or fewer runs	digest (Rebguns)

Sensing (Ch. 10; G7)

Idea	Source
Three presets: "design spec" (Heil, sub-inch within 6 ft), "Maxelbot" (0.6–0.7 in, 2.4–3.5 m, 8.33/N Hz) and "pessimistic" (Heil: σ 3.5 in, 14.5° bearing, drops past 0.6–0.9 m, cut-off at 6 ft)	digest (Heil, Maxim)
Noise on each range, not on the position, so geometric dilution at distance emerges by itself	digest (Kunkel)
Couple sensing to attitude: a tilting quad tilts its sound plane; receivers outside it drop out	digest (Heil)
Speed-of-sound calibration parameter (matters most underwater)	digest (Heil)
Relaying out-of-range positions with a hop-count "doubt level" (Heil App. B)	digest (Heil)
Cheap baseline: sector-only bearing (±30°) against full trilateration	digest (Heil)
The 50-sample running average adds lag that matters for kinetic theory and Daedalus	digest (Heil)
Drops: 1% acoustic and 0.2% RF; multiplicative U[0.9, 1.1] noise	digest (Maxim)
3D receiver: a fourth at (0, 0, d), giving $z = (b^2 - e^2 + d^2)/2d$ (our extension)	digest (Heil, Maxim)

Sound, BPP and the app

Idea	Source
Exaggerated pitch: stretch pitch deviations ~3× on the `emphasis` slider so a climb or hard turn becomes an audible swoop	chat (not built)
Event sounds: a knock and motor wind-down on a crash; a short chirp on launch and landing	chat (not built)
Info dock pane in BPP instead of the status line	chat (open item)
Port the Uniform Coverage fixes to the Android app (±15° cone; dead-end retry)	plan
FIXES item 27 (precompiled-header build bug): documented, not fixed	chat (open item)

The plan's capstone mission is now Section 6, extended to ground, surface and underwater vehicles.

6. The capstone: one mission, five kinds of vehicle

Ch. 13's central point is that physicomimetics only decides where each agent should go, and each vehicle's own controller works out how to get there. The book's experiments span several vehicle types (Maxelbots and Pioneers on the ground, surface boats, fixed-wing aircraft in simulation), but never all of them in one study. Running the same force laws on five kinds of vehicle, in one mission, makes the claim that the method is not vehicle-specific directly.

The plan's original capstone was drones only: launch in thirds, a perfect 3D lattice by compass and address, scouts probing a city, fluxotaxis finding a plume's source and height, and a chain relay back to the launch point. The extended mission keeps all of that and adds the other vehicles.

The mission: a harbour incident

A leak from a ship releases a chemical into both the air and the water.

Vehicle	Its part in the mission	Book chapters
Quadcopters	Launch in thirds, form a perfect 3D lattice, trace the airborne plume to its source and height	4.1, 8–9
Fixed-wing aircraft	Circle over targets for wide-area surveillance, since they cannot hover	13
Ground robots	Scouts test the dock area first; a chain relay goes into a warehouse or storm drain; robots stuck in the drain are led out as a chain (§7); a kinetic-theory sweep covers the sheds	15, 12, 7, ★§7
Surface boats	Map the slick or the bioluminescence; find the darkest route for a covert approach	6
Underwater vehicles	Trace the plume underwater; hold a relay chain from the surface down to depth	8–9, 12

Across media, the surface boats relay between sound underwater and radio above it. Ch. 16's typed force laws let different vehicle types share one formation, and depth bands work for underwater vehicles the way flight levels work for the drones.

New tasks

Task	Needs	Effort
★ **Cross-vehicle benchmark** (started, `BENCHMARK.md` : quadcopters, ground robots, boats and the ideal particles, tasks 1–2; F68): the same task with the same force-law settings on all five kinds of vehicle (quadcopter, fixed-wing, ground robot, surface boat, underwater vehicle), reporting the Ch. 16.9 measures for each: settling time, lattice quality, losses, and the gap between suggested and actual position	G18, G10, G11, G12	medium
★ Benchmark task 1: formation (2D lattice; 3D for the vehicles that can) (done for four vehicles)	G18	easy per vehicle
★ Benchmark task 2: goal through obstacles (Ch. 3.6) (done for four vehicles)	G18	easy per vehicle
★ Benchmark task 3: uniform coverage (Ch. 11)	G18, G4	medium
★ Benchmark task 4: chain formation (Ch. 12)	G18, G8	medium
★ ROV-style underwater vehicle (holonomic, hovers)	G11, G19	medium
★ Torpedo-shaped AUV (cannot hover or move sideways; heading particle)	G11, G19, G18	medium–hard
★ Underwater formation in depth bands (the counterpart of flight levels)	G11, G19, G20	medium
★ Underwater plume tracing, the plume carried by a current	G14, G19	hard
★ Relay chain from the surface down to depth, over sound	G20, Ch. 12	medium
★ Mixed-vehicle formations with typed force laws (Ch. 16)	G9, G18	medium
★ Surface boats as sound-to-radio gateways linking underwater and air swarms	G20, G15	medium
★ Harbour features of the world generator, and the full mission script	G21, everything above	hard

The benchmark matrix is four tasks × five kinds of vehicle, i.e. 20 runs, each over several seeds. That table is the scientific result; the mission demonstrates it.

The world generator (G21)

Our own generator, designed around this mission and serving every chapter. It is 3D and modelled on how real places are built, in general rather than any particular place. **Nothing is taken from the old map makers** in the CPT code (`MakeMap.c` for Textron, `MapGen.c`); `lib/world.lua` only *writes* their file format (`writeCPT`), so Zarzhitsky's PlumeGen and TraceGen can run on our worlds as an independent check of our plume port.

Approach (decided 2026-10-17): procedural, calibrated to published statistics of real places. Importing real places (OpenStreetMap footprints with `building:levels`, elevation models, port depth charts) is future work; it would then also calibrate the statistics.

Layers, in build order:

Layer	What	Based on
Terrain (**done**, `lib/terrain.lua`)	Heightfield: coastal plain, hills on it fading toward the shore, river valleys; one rule keeps every dry slope drivable (≤ 15% along the grid) and also shapes valley walls and banks	Fractal value noise; meanders by the disturbed periodic model (Ferguson 1975, 1976), with neck cutoffs: sinuosity 1.4, wavelength 11 widths (Leopold and Wolman 1960), irregular, as natural meanders are (F64)
Water (**done**, `lib/terrain.lua`)	Sea, harbour, rivers, ponds with depth; dredged approach, channels, turning basin, berths with 1:3 sides, quay walls vertical; shelf shoaling to the shore	Los Angeles main channel 53 ft (16.2 m); Long Beach approach 76 ft (23.2 m): ours 24 m and 16 m
Zones (**done**, `lib/town.lua`)	Districts around centres, density falling exponentially outward, typed by Local Climate Zone (compact/open high-, mid-, low-rise, large low-rise, sparsely built, heavy industry; trees, fields, paved, soil, water); districts for ports, parks, industry; woods a stated share of open land	Clark (1951); Stewart and Oke (2012), building fraction: compact high-rise 40–60%, > 25 m; open low-rise 20–40%, 3–10 m; sparsely built 10–20% (corrected, F61)
Streets (**done**, `lib/town.lua`)	Arterials through everything, grids in cores, curving collectors and cul-de-sacs in suburbs, winding rural roads; bridges; every street connected to the arterials	Manhattan 1811: streets 60 ft every ~264 ft, avenues 100 ft every ~900 ft (80 × 274 m); Chicago 66 ft streets, 100 × 200 m; fire code: dead ends to 750 ft, 96 ft turnaround
Lots, buildings (**done**, `lib/buildings.lua`)	Frontage lots with setbacks; rectangles and L, T, U footprints along streets; whole floors (3.5 m), heavy-tailed, lower toward the edge; farmsteads; the gap rule; each zone's building fraction and mean height inside the published LCZ ranges	Schläpfer et al. (2015): one-storey buildings peak at 3.5 m, heavy tail, height falls from the centre; US median new lot 8,400 sq ft (2023); Stewart and Oke (2012) ranges (F62)
Vegetation (**done**, `lib/vegetation.lua`)	Woods of trees over zone A (300 a hectare, closed above and open below the crowns, so drones fly between the trunks; F70), hedgerows on field boundaries, street trees, garden and park trees (trunk and raised crown); trees block sight and sound, are porous to air (for the plume, later)	LCZ land covers; canopy 20 m, mature woodland 200-400 trunks a hectare, street trees 12 m apart, fields 250 m
Mission features (**done**, `lib/features.lua`)	Quay, piers, ship at a berth (the leak), warehouses with doorways, container yard with pockets, storm drain and outfall, launch pads	The capstone; 20 ft containers (6.1 × 2.44 × 2.59 m)

Presets (parameter tables for one generator): **rural** (fields, woods, hedgerows, farmsteads, a river and pond, winding roads, rolling terrain); **suburban** (houses on lots, curving streets, cul-de-sacs, trees, a park with a pond); **city centre** (compact mid- and high-rise grid, heights falling from the centre, plazas, a park); **industrial port** (warehouses, container yard with pockets, quays, dredged basins); **harbour town**, the capstone (all of these around a harbour with real depth structure, the ship at its berth, storm drains and outfall, launch pads).

Everywhere: gaps between buildings fit an agent or are closed (the Android port's rule); worlds are seeded from their own stream, so a seed always gives the same world and sweeps vary one knob at a time; distant districts are merged into static meshes, with full detail only in the mission area.

Outputs (done): BPP geometry (terrain, water and seabed, buildings, trees, the drain's manholes); grids of ground height, water depth, zone and occupancy at a chosen height (`raster`, `writeRaster`; `writeCPT` for Zarzhitsky's

tools); line of sight (`sight`); metric masks (cells, shadow cells, `traps`, water). **Later:** wind with height and currents.

The book's worlds stay as the book draws them, as presets for the chapter ports: Ch. 3's 12 discs, Ch. 4's coverage box, Ch. 7's corridor (Fig. 7.2's shapes), later Ch. 11's environments, Ch. 12's maze and Ch. 14's field. The core and these presets came in M0 (`lib/world.lua`, `Worlds.lua`); the realistic layers follow in the table's order, one patch each: terrain and water are done (`lib/terrain.lua`; the five presets in `lib/places.lua`, shown by `Worlds.lua` presets 5–9; `checks/terrain.lua`), zones and streets (`lib/town.lua`, `checks/town.lua`), lots and buildings (`lib/buildings.lua`, `checks/buildings.lua`), and the mission features and vegetation (`lib/features.lua`, `lib/vegetation.lua`, `checks/features.lua`). All five presets are complete (FINDINGS F60–F63).

Order

Build G18 and the cross-vehicle benchmark early, right after the Ch. 13 fixes, starting with ground robots and surface boats because they are simpler than underwater vehicles. That tests the claim cheaply. The full mission comes last, since it needs nearly everything else.

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7. Escaping cul-de-sacs: the chain fountain

The gap in the book. Lennard-Jones swarms flow round obstacles, but agents caught in a concave pocket stay stuck. Ch. 14 leaves this as future work and suggests that stuck robots switch to the kinetic-theory gas model to escape (§14.11, PDF 503). Hettiarachchi's thesis also lists cul-de-sacs as an open problem and never defines "stuck".

The idea. `demo/WyomingWill/ChainFountain/chain-fountain.lua` simulates a bead chain pouring out of a jar (the Mould effect). The part of the chain already outside pulls the rest over the rim; links are picked up off the pile one at a time, and because a joint can only bend so far (30° in the exhibit), the chain arches up and over the rim instead of dragging across it. The same mechanism can lead stuck agents out of a cul-de-sac, as a chain behind an agent that is not stuck. This is an alternative to the book's chain formations (Ch. 4.2, 12), which deploy a relay outward from a base; this chain pulls agents out of a trap.

Chain fountain	Escaping a cul-de-sac
The jar	The cul-de-sac
The end hanging outside	A free agent (not stuck) as the head
Gravity on the falling chain	The goal force, felt only by agents already outside
Links picked up one at a time	Stuck agents join the chain's tail in order, nearest first
Fixed link length; joints bend at most 30°	Each agent holds a set distance behind the one ahead, with a limit on the angle between successive links
The arch over the rim	Clearance at the trap's mouth

Why it might work well:

- **The pull grows as agents get out.** In the fountain the weight of the falling chain drives the flow, so the more is outside, the stronger the pull. If only agents outside the trap feel the goal force, the swarm behaves the same way.
- **The bend limit stops corner-cutting.** Plain follow-the-leader cuts corners and drags followers into walls; a bend limit makes the chain swing wide round the mouth.
- **It suits real vehicles.** A bend limit is in effect a minimum turn radius, which ground robots and fixed-wing aircraft have anyway (Ch. 13). In 3D, drones can do what the bead chain does: arch up and over the wall rather than thread back out through the mouth.

What the fountain's latest runs add. The arrangement of the pile decides the arch. Poured as a jumbled heap, as in the experiments, a ball chain stands at h_2/h_1 = 0.11–0.15 in Bullet against 0.12–0.14 measured (Biggins and Warner

2014; Pantaleone 2019). Laid as the exhibit's tidy coil, the same chain is picked up at the wall, rides the rim and stands at only 0.06 (commits `1aa427a`, `f938b01`). For the swarm this suggests two things:

- **The order of pick-up matters.** If stuck agents lie along the trap's wall and the nearest one always joins the tail, the chain will ride the wall, like the coil riding the rim. Recruiting from inside the cluster, away from the walls, may give the clearance the heap gives.
- **Only the agents near the pick-up need to act.** The fountain now simulates only the links within 4 cm of the pick-up and keeps the rest of the heap fixed, and still matches the full heap. In the same way, only stuck agents near the chain's tail need to take part, so the method stays local as the swarm grows.

What doesn't carry over. The fountain's push comes from the pile pushing back on each link as it is lifted. Agents have no pile, so the arch height will not follow Biggins and Warner's relation for h_2. Whether a virtual version of that push helps is an experiment, not a given.

Tasks

Task	Needs	Effort
★ **Cul-de-sac world:** U-shaped and deeper pocket obstacles added to the Ch. 14 obstacle field, and a definition of "stuck" (no progress toward the goal over a time window)	G13	easy
★ **Virtual chain law:** set distance to the agent ahead, bend limit, recruitment of the nearest stuck agent at the tail, goal force only outside the trap. 2D particles first	G9	medium
★ **Compare escapes:** plain Lennard-Jones (baseline), kinetic-theory escape (Ch. 14.7), the chain-fountain escape, and the Ch. 12 chain law. Measures: share reaching the goal, time, collisions, agents out per second, clearance at the mouth	G3, G16	medium
★ **Sweeps:** link length, bend limit and head speed; whether the outflow rate settles to a steady value as the fountain's does	G3	easy
★ **Pick-up order:** stuck agents strung along the wall against clustered inside the pocket, and recruiting the nearest agent against one away from the walls; measure clearance at the mouth and wall contacts (the coil-versus-heap result)	G3, G8	easy
★ **Vehicles:** ground robots round corners; quadcopters arching over the wall in 3D; fixed-wing aircraft, whose turn radius is the bend limit	G10, G12, G18	medium
★ **In the capstone:** ground robots stuck in the docks or the storm drain, led out by a free one (§6)	§6	—

8. Build order

This revises the plan's order using what the book says.

Before porting a task, read its chapter note, read its `.nlogo` in `netlogo/`, examine the Android app's version (mostly built from the `.nlogo`; sometimes improved, sometimes wrong), and read the matching thesis digest. The book's chapters leave out values that the models and theses contain. `WORKFLOW.md` describes the steps every task goes through.

Step	Work	Why now
0	**G2 shared RNG** (move the existing multiply-with-carry generator out of `Uniform_Coverage.lua`, add a seed setting) **+ G3 batch runner + G1 monitors + G21 world generator** (core and the first book presets), then Ch. 2's three tests	Every measurement and every world after this depends on them
1	**Ch. 13 on the drones:** η → adaptive friction → static friction → adaptive cap, each switchable, measured with the Ch. 16.9 metrics on Newtonian 2D/3D over several seeds	The plan's step 1, now with four parts and acceptance metrics
2	★ **G18 vehicle interface + cross-vehicle benchmark**, first with ground robots (G10) and surface boats (G11) alongside the quadcopters: formation and goal through obstacles (§6). **Done (M2):** the claim holds for the holonomic vehicles; ground robots fail only through the minimum wheel speed (`BENCHMARK.md`, F68)	Tests the vehicle-independence claim early and cheaply
3	Ch. 3 fidelity checks (G_t sweep incl. 3D, Disable One, Hooke, honeycomb) and Ch. 4.3/Ch. 11 coverage against Tables 4.2, 11.1 and Fig. 4.19	Cheap; firms up what is already "done"
4	Ch. 4.1 perfect square → cube, then the triangle; Ch. 17 Method 2 as the alternative	The plan's step 2
5	G4 velocity commands + G5 point mass; Ch. 4.4 Couette and Ch. 7 theory checks	Opens all the kinetic-theory and quark work
6	Ch. 14 offline world → Ch. 15 scouts (particles), then Daedalus; add cul-de-sacs and compare escapes: kinetic theory (Ch. 14.7) against the ★ chain fountain (§7)	The plan's step 3, in the order the book's dependencies need; the cul-de-sac is the book's open problem
7	Ch. 19 formation APO with G6 fields → ★ from altitude, moving hotspot	The plan's step 4
8	Ch. 5 LOPF and Ch. 16 type table: square, line, star, V, rings, protection	New, cheap on the formation code
9	G7/G8/G10 sensing and ground vehicles → Ch. 10 checks, Ch. 11 and 12 replicas, Ch. 12 chains, then ★ 3D shafts	Hardware-faithful track
10	Ch. 8 unit tests → G14 plume → Ch. 9 → ★ fluxotaxis 3D	Flagship
11	G11 ASV → Ch. 6 → ★ ROV and AUV (G19, G20); G12 fixed-wing → Ch. 13.7–13.12; complete the cross-vehicle benchmark with all five kinds of vehicle; Ch. 1 demos; Ch. 18 benchmark	Remaining vehicles and chapters
12	★ **Capstone:** the world generator's harbour (G21), underwater plume and relay, mixed-vehicle formations, the full mission (§6)	Needs nearly everything else